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RESEARCH ARTICLE



An ontology-based data access framework for enhanced sewer network management

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ABSTRACT

Effective urban water network management is often hindered by data heterogeneity and a severe lack of interoperability. This study leverages the power of the Semantic Web to introduce a new application ontology, the SewerNetworkOntology, coupled to an Ontology-Based Data Access (OBDA) framework specifically designed for sewer network management. Our framework integrates a specifically designed conceptual layer, representing the physical structures of these networks, linked to a data layer through formalised mappings. A key contribution is the framework's ability to automatically identify data inconsistencies that remain hidden with traditional methods. We demonstrate the effectiveness of our system through its application to the wastewater and stormwater networks of Montpellier Métropole.

ARTICLE HISTORY

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1. Introduction

Climate change and urbanisation are placing increasing stress on the infrastructure, posing a challenge for upholding the services (Fontecha *et al.* 2020, Hughes *et al.* 2021). Among these assets, sewer systems are a key component of urban infrastructure supporting the daily needs of residents. When the conditions for which these systems were originally planned change, from extreme weather to urban sprawl, they struggle to provide adequate services (Hughes *et al.* 2021) placing a strain on the operation of the urban infrastructure (Araya and Vasquez 2022). This leads to a higher risk of pluvial flooding aggravating the release of polluted stormwater into the environment (Mikovits *et al.* 2018). To reduce long-term costs and limit disruptions to city life and the environment, strong planning and proactive management are essential (Haydar *et al.* 2026). However, lack and incompleteness of data and terminological ambiguity (Ceccaroni *et al.* 2000) hinder the ability to assess the situation and properly maintain and manage the networks. Khaleghian *et al.* found that data quality varies across municipalities, with 11–35% of sewer inspection records exhibiting issues that must be addressed to support effective asset management strategies (Khaleghian and Shan 2023). Ultimately, the quality of condition assessment data directly shapes the reliability of decisions made for sewer infrastructure management.

Ontology-Based Data Access (OBDA) frameworks offer a potentially effective solution facilitating the integration and data access by leveraging ontologies to provide a unified semantic view of the data (Xiao *et al.* 2018). This can lead to improved communication, information exchange and collective learning within the domain. Moreover, OBDA enables domain experts to formulate queries using ontology terminology hiding the complexity of the actual data structure (Kharlamov *et al.* 2017). Several attempts have been made to provide ontology-based systems in wastewater and stormwater domain (Ceccaroni *et al.* 2000, Luccarini *et al.* 2014, Zeb 2020) focusing either on utility infrastructures or wastewater treatment approaches. While significant progress has been made in the development of ontology-based systems tackling wastewater and stormwater, there is a notable absence of a framework tailored specifically to manage their networks.

We present an OBDA framework for sewer networks, focusing on the automated detection of inconsistencies in the underlying data sources. The system leverages an application ontology built for sewer

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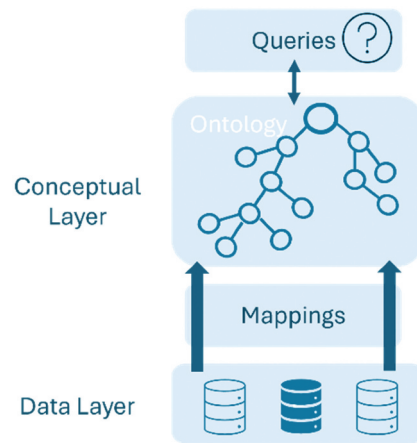


Figure 1. OBDA general framework.

networks to provide a unified and coherent view of the data, which is crucial in domains where data originates from diverse sources.

2. Methodology

To address the evolving challenges of data access for urban water management, availability, heterogeneity and quality, the paper proposes an OBDA system, a *semantic paradigm* (Xiao *et al.* 2018) that allows the access of data through a conceptual layer. The OBDA framework is logically composed of three layers as shown in Figure 1:

- The conceptual layer is defined by a new application ontology specifically designed to model the physical structures and operational aspects of urban water networks. The ontology integrates key concepts of the domain using the INSPIRE European directive (2007) and the RAEPa French Geostandard (2019). Protégé (5.6.1) (Musen 2015), an ontology editor and a freely available open-source tool developed by Stanford University, is used to build the ontology in OWL 2 QL (Motik *et al.* 2012) format as it is one of the most widely used ontology development tools (Escórcio and Cardoso 2007).
- The data layer includes the various sources (typically stored in relational databases) describing the wastewater and stormwater networks.
- The mappings link the terms of the conceptual layer to the data sources.

3. Results and discussion

The SewerNetworkOntology establishes a hierarchical structure of classes such as the *NetworkElement* class that is subdivided into *Pipe* and *Node* classes, with the latter further specialising into *Appurtenance* and *Structure* classes. The ontology models different sewer network types represented as classes: *StormwaterNetwork*, *WastewaterNetwork* and *CombinedNetwork*. It also includes an *Event* class, specialising into *Maintenance*, *Inspection* and *Incident*. Other classes such as *Material*, *PipeCategory* and *PipeCirculationMode*, featured as code lists or enumerations in the standards, enrich the data classes like *NetworkElement*. An excerpt of the class hierarchy of the ontology is shown in Figure 2. The ontology consists of about 53 classes, 18 object properties, 39 data properties and more than 500 axioms between logical and declaration axioms.

The data layer describes the wastewater and stormwater networks of Montpellier Metropole. These sources consist of GIS files for pipes, manholes and structures, totalling tens of thousands of features. They are available in open access on the Metropole's open data portal.

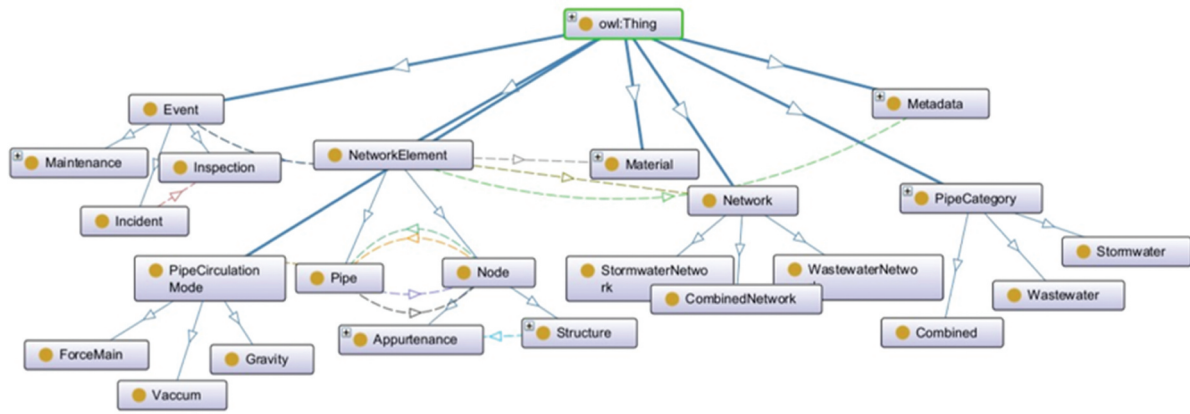


Figure 2. Excerpt of the class hierarchy of the conceptual layer developed for this study.

The mappings allow the extraction of new information from the database that is not explicitly available in the data and is necessary for the management of the network. Using the geometry of the pipe and nodes extracted from the shapefiles and utilising the spatial functions in PostGIS *ST_StartPoint*, *ST_EndPoint* and *ST_Geometry*, the nodes located at the end of the pipes are identified. In the ontology, the object properties *hasInitialNode* and *hasTerminalNode* are introduced to describe the connection between a pipe and the nodes at its endpoint.

In our implementation, we use PostGIS in combination with QGIS to store the data, Ontop (5.1.1) (Calvanese *et al.* 2016) as Protégé plugin to reformulates the SPARQL (Harris *et al.* 2013) queries into SQL, and the HermiT (Shearer *et al.* 2008) reasoner to check the logical consistency of the ontology. Two versions of the data accessed on July 2023 (v1) and February 2024 (v2) are used to identify the effect of updating the dataset.

The framework's primary utility is its ability to identify inconsistencies that emerge from the integration of heterogeneous data. In sewer networks, information is often stored across multiple databases maintained by different departments. These records may contain inconsistencies due to variations in data entry practices or updates made over time.

By leveraging ontological principles, data can be organised in a common framework as it grows and evolves limiting discrepancies and conflicting information. Ontology-based systems also provide a tool to query and analyse data repositories to detect and rectify inconsistencies. Considering the sheer volume of data, manual inspection for discrepancies becomes essentially impractical.

We defined a set of SPARQL and GeoSPARQL (Car *et al.* 2024) queries that are formulated to detect critical inconsistencies, including topological errors and violations of hydraulic norms, e.g., pipes with missing connections and inconsistent flow direction. The execution of these queries allowed the evaluation of the integrity and accuracy of the frequently updated Montpellier Métropole open data. The data accessed on July 2023 (v1) and February 2024 (v2) are assessed to identify the source of the inconsistencies and the effect of updating the dataset. Table 1 shows the evolution of the network in terms of the number of the network elements after 7 months. The growth in the nodes and pipes numbers in the recent database could be linked to network expansion, data correction or enrichment.

The shapefiles of the two database versions were imported into PostGIS and queried to detect the inconsistencies and modifications and their effect on the network. Using the developed OBDA system, the process for querying the recent dataset (v2) was efficient and straightforward as nothing had to be changed in the mappings and queries. To see the effect of modification on data completeness, SPARQL queries were executed to check for instances of a specific class within the dataset that lacks an important attribute. For

Table 1. Pipe and nodes in the two versions of the database.

Source	Number of Pipes	Number of Nodes
Database v1	51,013	54,955
Database v2	51,368	55,431

Based on the results of Q3 and Q4, redundant information increased after the updates indicating multiple entries of the same network entities. The redundant information in the database has different IDs making their manual detection hard over the network. The multiple data entries might stem from the procedural aspect of the update process over time.

As for Q5, the number of missing connections decreased after the update. This query deals with the pipes with missing nodes. Each pipe should have an initial and a terminal node. Using the object properties *has InitialNode* and *has TerminalNode* in the ontology, the query selects pipes where either the initial node or the terminal node is not assigned a value, indicating missing or incomplete data. The pipes retrieved can be either pipes with no nodes or pipes missing one of the end nodes. The result of this query is related to Q1 where some nodes are mislocated and do not fall on the corresponding pipe geometry, so they appear as isolated nodes. Another reason that led to the decrease in the number of pipes with missing nodes could be the addition of nodes, which also supports the increase in the number of nodes in v2. Thus, missing nodes can be the result of digitisation errors and missing information.

The results of Q6 which detects the pumping station that are connected to gravity pipes instead of forced pipes, are the same in both versions, indicating that the update didn't affect the pipes connected to the pumping station. Pumping stations serve as the initial nodes of forced pipes and are placed where gravity flow is insufficient to maintain pressure and ensure the transport of the fluid. With this query, all pumping stations that are the initial nodes of gravity pipes are returned, indicating that either the pumping station is misplaced, out of order, inaccurately documented or the flow direction is digitised incorrectly and the pumping station is in fact a terminal node.

However, some of these inconsistencies are resolved while others are added in the new version of the dataset. Due to flow direction inaccuracy, the pumping station in some results is a terminal node instead of an initial node as seen in Figure 4 where the arrow direction shows that previously the pumping station (cornier des oiseaux) had two inflows (Figure 4(a)) and after the update one inflow and one outflow (Figure 4(b)).

For the result of Q7, the inaccuracy in the pipe's flow direction in v2 slightly grew. Two sources were discovered for this inaccuracy. The inconsistency in flow direction is retrieved based on the elevation values where for gravitational pipes, the flow direction is digitised in the shapefiles from upstream nodes to downstream nodes. This indicates that in a gravity-fed flow, the upstream elevation should be higher than the downstream elevation to propel the flow. One case was found to be a data entry error in the upstream elevation, where there is a huge difference from the downstream elevation. The other source of inaccuracy is a different version of data input error.

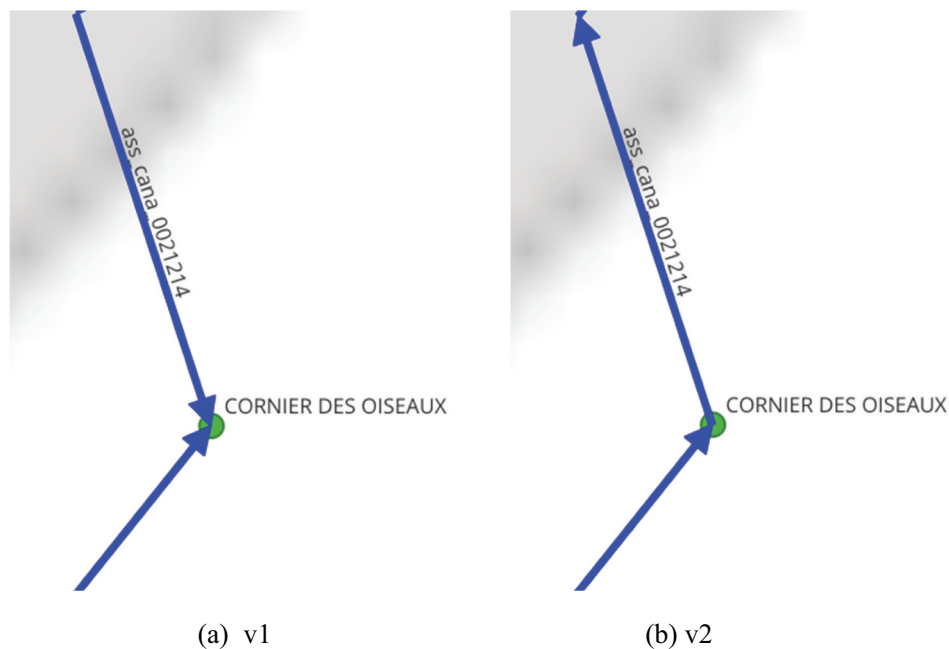


Figure 4. Example of a pumping station connection with the green dot represents the station, and the arrowed blue lines represent the flow.

For Q8, the number of pipes with diameters less than 200 mm has decreased after the update. The pipe should have a nominal diameter equal to or greater than 200 mm, including private networks, according to the technical guide of Montpellier Métropole. This simple query retrieves pipes with a diameter smaller than 200 mm which may hint at a data entry error. Similar to previous results, the update resolved some data entry issues and added others. Still, for this query, we can't consider all results to be regulatory violations. Simply, they could be ancient pipes laid before the regulations. However, the fact that some pipe diameters were changed after the update means that there could be pipes among the results with a data entry error, where 22 pipes with diameters less than 200 mm were added to v2, while another 64 pipes were removed.

In summary, we confirm that missing data is a common problem in urban water networks, which may lead to over- or underestimation of flood volume and duration (Chahinian *et al.* 2019, Belghaddar *et al.* 2021, Shrestha *et al.* 2022). In a context of global change, climatic forcing is expected to evolve, and network managers will have to face new adaptation measures (Yu *et al.* 2022). Thus, missing or inconsistent data on the network's current geometry will add uncertainty to the adequacy of the predicted measures. Let us note that the examples considered for the different queries may not represent all results. Other sources could be behind the discrepancies in the data such as data processing errors. The sources of the inconsistencies of the analysed instances are confined to two main categories: data collection error or data input error.

Our discussion provides a brief view of the effect of updating the data using a limited number of queries for different forms of inaccuracies and inconsistencies. Nevertheless, our OBDA approach provides managers with a tool to track the updates' impact ensuring the accuracy and reliability of the published data without the need to know how the data is represented in the databases. It allows the detection of unperceived discrepancies reducing the time on manual inspections, especially for buried networks where malfunctions go unnoticed. The executed queries provide an initial step towards defining a set of criteria that can help managers identify inconsistent information.

4. Conclusions and perspectives

In this framework, we propose an Ontology-Based Data Access (OBDA) system allowing data on sewer networks to be accessed and queried through a semantic layer that defines controlled vocabulary. Our ontology is inspired by the RAEPA and the INSPIRE standards and validated by domain experts. A mapping is created between the ontology terms and the data sources of Montpellier Métropole using the OBDA reasoner Ontop. The advantage of the adopted framework lies in the fact that additional knowledge may be acquired without modifying the raw data. GeoSPARQL and SPARQL queries are executed over the ontology layer allowing, e.g., to detect inconsistencies. The effect of two data updates was analysed. While the updated data resolved some inconsistencies, it created others. Most of the inconsistencies are traced back to errors in data collection and data input.

The challenge of the system is that some SPARQL and GeoSPARQL query features are not supported in Ontop. As for the ontology, it encapsulates the core concepts of the domain in a flexible declarative structure, enabling future growth and expansion. Spatial and non-spatial queries can be added to provide a more comprehensive data integrity evaluation. Additionally, the system can be used for different sewer networks by only updating the mappings between the ontology and the new data sources.

The OBDA framework's first version is accessible on the GitHub repository <https://github.com/batoulhaydar/OBDA-Sewer-Network> for public use with the ontology, mappings and queries defined in this paper and a link to the public data used.

Author contributions

CRediT: **Batoul Haydar:** Conceptualization, Formal analysis, Methodology, Software, Validation, Writing – original draft, Writing – review & editing; **Claude Pasquier:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing; **Umberto Straccia:** Conceptualization, Formal analysis, Methodology, Validation, Writing – original draft, Writing – review & editing; **Nanée Chahinian:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing.

Disclosure statement

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